DEVLEARN 2026 / SESSION 606 / THE BACKUP LEAGUE

THE MISSION ARCHITECT COMPLETE TOOLKIT

Eight resources. Four stages. One playable learning goal.

Start with the Field Guide, then follow the stages that fit your project.
The numbered route is a guide, not a mandatory sequence.

PLAN YOUR MISSION

01 Mission Architect's Field Guide page 2

02 Five-Step Micro-Escape Design Canvas page 4

03 Pilot Proposal page 6

BUILD YOUR CHALLENGE

04 Puzzle Mechanics + Creative Reveals page 7

05 AI-Assisted Puzzle Design page 17

TEST & LAUNCH

06 Playtesting & Puzzle Validation page 18

07 Logistics Planning page 19

MAKE THE LEARNING STICK

08 Debrief & Learning Transfer Guide page 20

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THE MISSION ARCHITECT'S FIELD GUIDE

Start with performance. Make the thinking playable.



FIVE STEPS FROM WORKPLACE NEED TO LEARNING GAME

01	**FIND THE PERFORMANCE GAP**	Where are people struggling, and is practice an appropriate response? **OUTPUT: A behavior worth practicing**
02	**MAKE SUCCESS OBSERVABLE**	Given which information, what action will learners take, and what counts as correct? **OUTPUT: A measurable objective and success criteria**
03	**CHOOSE MECHANIC + STORY**	Choose an action that requires the target thinking, then add role, problem, and stakes. **OUTPUT: A playable concept and meaningful story**
04	**BUILD THE CHALLENGE**	Create the evidence, rules, artifacts, and answer key; test with a fresh solver. **OUTPUT: A playable, verified prototype**
05	**DESIGN FEEDBACK + TRANSFER**	Plan self-checks, hints, debrief, recovery, and a new situation to test the skill. **OUTPUT: A feedback and transfer plan**

BACKUP LEAGUE EXAMPLE: MEMORY MELTDOWN

Players examine six fictional incident reports, identify the three highest-priority emergencies, and sequence them using risk and timing criteria. Correctly arranged card backs reveal a verification message, and successfully completing the chapter earns a star-chart fragment for the final rescue puzzle.

DESIGN RULE: Could someone win without using the target behavior? If yes, redesign the interaction. A theme supplies purpose and stakes. The mechanic should require the same thinking that learners need beyond the game.

YOUR FIRST MISSION / QUICK DESIGN CANVAS

Use these four questions to sketch a five-minute prototype or start a fuller design canvas.

TASK

What decision or skill must people practice?

SUCCESS

What would observable success look like?

MECHANIC

What must players sort, compare, sequence, route, or assemble?

FEEDBACK

How will players get help and understand why they are right?

THE FIVE-MINUTE NOVICE TEST

- ☐ Can someone start without an oral explanation from the designer?
- ☐ Can a player reach and justify the intended solution using only the supplied evidence?
- ☐ Can they self-check, request a hint, and recover in time?
- ☐ Do instructions stay readable and usable with the actual room, group, and devices?
- ☐ Can you ask a new scenario or debrief question to check learning beyond the activity?

END WITH A QUICK DEBRIEF

Don't stop at revealing the correct answer. Ask players to explain their reasoning and connect the experience to a real-world situation. Try asking: "What evidence guided your decision? Why did you choose that approach? Where could you apply similar thinking outside the game?"

AI ATTRIBUTION Artwork generated with OpenAI tools; AI-assisted text reviewed and edited by the author.

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FIVE-STEP MICRO-ESCAPE DESIGN CANVAS



An editable worksheet for one practice-worthy decision.

Start small. Complete the steps in order, then revisit earlier choices when a test reveals a problem.

01 FIND THE PERFORMANCE GAP

Observed difficulty: What should happen, what happens instead, and what evidence suggests why?

02 MAKE SUCCESS OBSERVABLE

Given [situation/materials], learners will [observable action] to [measurable standard].

03 CHOOSE MECHANIC + STORY

Which player action requires the target thinking? What role, problem, and stakes give that action purpose?

04 BUILD THE CHALLENGE

What evidence, constraints, artifacts, and intended answer will you prototype and test?

05 DESIGN FEEDBACK + TRANSFER

How can players check/learn why, get hints, recover, and use the skill in a new situation?

DELIVERABLE: A performance need, success criteria, playable concept, tested prototype, and feedback plan.

PUZZLE BLUEPRINT / PROTOTYPE WORKSHEET

Make a small packet, test it, and revise before adding polish.

MAP THE ACTUAL PLAYER EXPERIENCE

Briefing: What is the player's role, problem, and goal?

Inputs: Which clues, reference criteria, or artifacts do they receive?

Action: What must they decide, sort, assemble, or verify?

Answer / rationale: What solution follows from the supplied evidence?

Progress / reveal: What satisfying reward follows success? Could a player earn it without using the target skill?

Feedback / rationale: How will players know they are right and understand the reasoning?

Hints / recovery: What happens when they stall or run out of time?

Transfer: What new scenario or question will reveal independent understanding?

SKETCH THE CHALLENGE LOOP

Challenge–Action–Reward lens: Rachel Arpin (ATD, 2023).

CHALLENGE	ACTION	REWARD
A problem with meaningful stakes	A decision requiring the target skill	Clear resolution and useful feedback

AI ATTRIBUTION Artwork generated with OpenAI tools; AI-assisted text reviewed and edited by the author.

PILOT PROPOSAL

A small, measurable request with human-validated design.



MAKE THE CASE FOR A SHORT PILOT

Pitch the performance decision and evidence, not "an escape room." Stakeholders typically care about:

- ☐ Problem and evidence - what has been observed?
- ☐ Learner behavior and success standard - what must people decide or do?
- ☐ Pilot audience, duration, location, and materials
- ☐ How you will check decisions and individual reasoning
- ☐ What comparable task could reveal later transfer?
- ☐ Pilot decision rule: What evidence would justify expanding, revising, or stopping the pilot?
- ☐ Approval needed and proposed next step(s)

REQUEST PLANNING:

Jot down your thoughts below, so you can succinctly phrase the request.

WHY THIS PROBLEM?

Name the inconsistent decision, errors, delays, or risk.

WHY THIS FORMAT?

Explain why people need practice applying criteria.

WHAT WILL IT TAKE?

Scope a short pilot, materials, staffing, and novice playtest.

HOW WILL WE KNOW?

Observe decisions and reasons; try a comparable case later.

EXAMPLE OF A PILOT PROPOSAL

"Suppose a needs assessment identifies difficulty prioritizing competing incidents. We could pilot a 10-minute, six-report sorting challenge with a small group, observe their decisions and reasoning, and use a comparable new scenario to determine whether the approach should be revised or expanded."

AI ATTRIBUTION Artwork generated with OpenAI tools; AI-assisted text reviewed and edited by the author.

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PUZZLE MECHANICS + CREATIVE REVEALS

A practical menu for making the right thinking playable.



Start with a workplace or learning decision, choose a game action that requires it, then add the reveal.
New to making puzzles? Start with a short sorting or sequencing challenge and add just one reveal.

LEARNING MECHANIC	REVEAL TECHNIQUE
What players must think through or decide: rank, diagnose, compare, sequence, and more.	How an answer produces an enjoyable moment: a message, movement, symbol, or key.

USE THE FIVE-STEP DESIGN SEQUENCE

01	02	03	04	05
Find the gap	**Define success**	**Choose mechanic + story**	**Build + test**	**Feedback + transfer**

MATCH THE LEARNING OBJECTIVE TO A FIRST MECHANIC

LEARNERS NEED TO...	POSSIBLE MECHANIC	EXAMPLE DECISION
Prioritize competing needs	Sort / rank	Triage alerts; justify the order.
Recognize categories or risks	Match / classify	Match cases to rules or services.
Follow a procedure	Sequence steps	Restore a safe or valid workflow.
Evaluate conflicting evidence	Evidence comparison	Reconcile logs and witness reports.
Diagnose an issue	Fault isolation	Rule out causes from test results.
Apply limits or tradeoffs	Routing / allocation	Stay within constraints.
Coordinate across roles	Split information	Combine facts held by teammates.
Synthesize several results	Meta-puzzle	Interpret artifacts from earlier chapters.

EXAMPLE FROM THE BACKUP LEAGUE Memory Meltdown combines sorting and sequencing with a card-back reveal. Players must select three priority incidents from six reports and arrange them correctly before receiving their star-chart fragment. The learning mechanic practices prioritization; the reveal rewards the completed decision.

HOW TO READ THE TECHNIQUE CARDS

LEARNING CONNECTION the behavior being practiced. **HOW IT WORKS** the action players take. **HOW TO MAKE IT** the materials and construction. **DESIGN CHECK** the question to validate the experience.

DECISIONS + EVIDENCE

01-06 | Choose evidence-based actions with defensible answers.

Use these as idea starters, then tailor the evidence, construction, and feedback to your learners.

01 SORTING & RANKING

LEARNING CONNECTION Triage, escalation, risk assessment, or comparing competing needs.

HOW IT WORKS Players rank or classify case cards using explicit criteria, then explain the decision.

HOW TO MAKE IT Print short scenarios, a criteria guide, a sorting surface, and an answer with rationale.

DESIGN CHECK Can someone justify each placement from the criteria, without guessing?

02 MATCHING & CLASSIFICATION

LEARNING CONNECTION Applying a policy, choosing resources, or recognizing types of situations.

HOW IT WORKS Players match realistic cases to the correct category, owner, or service.

HOW TO MAKE IT Prepare case cards, a reference guide, and several plausible options.

DESIGN CHECK Are the distractors fair, and can every match be explained?

03 SEQUENCING & DEPENDENCIES

LEARNING CONNECTION Following procedures, handoffs, safety steps, or workflow dependencies.

HOW IT WORKS Players order process tiles, taking account of rules or exceptions.

HOW TO MAKE IT Draft the ideal workflow, then shuffle steps and add meaningful constraints.

DESIGN CHECK Are alternate orders either valid or clearly ruled out?

04 EVIDENCE COMPARISON

LEARNING CONNECTION Critical reading, investigation, source evaluation, or risk review.

HOW IT WORKS Players reconcile statements, documents, logs, and conflicting claims.

HOW TO MAKE IT Create a small dossier with reliable and unreliable clues plus decision criteria.

DESIGN CHECK Can learners explain why one source outweighs another?

05 LOGIC GRID / DEDUCTION

LEARNING CONNECTION Systematic elimination and evidence-based reasoning.

HOW IT WORKS Players combine linked clues to determine a schedule or configuration.

HOW TO MAKE IT Draft a solution first; write clues and provide a printable logic grid.

DESIGN CHECK Prove a unique solution if only one is intended. Test independently.

06 ROUTING & CONSTRAINTS

LEARNING CONNECTION Scheduling, navigation, dependencies, equipment configuration.

HOW IT WORKS Players choose a route or arrangement that satisfies several restrictions.

HOW TO MAKE IT Provide a map or network, rules, prohibited routes, and a checking method.

DESIGN CHECK Verify every route against all rules; watch for accidental shortcuts.

FAIR REASONING Can a fresh player justify the answer using only the criteria provided?

SYSTEMS + FIRST REVEALS

07-12 | Move from diagnoses and tradeoffs to earned discoveries.

07 DIAGNOSIS / FAULT ISOLATION

LEARNING CONNECTION Troubleshooting, identifying causes, interpreting diagnostic results.

HOW IT WORKS Players examine symptoms, run or interpret tests, and narrow possible causes.

HOW TO MAKE IT Create a symptom report, test evidence, candidate faults, and a decision path.

DESIGN CHECK Does evidence determine the answer, not random trial and error?

08 RESOURCE ALLOCATION

LEARNING CONNECTION Tradeoffs, planning, prioritization, and justifying choices.

HOW IT WORKS Players distribute limited repair time, supplies, or staff across needs.

HOW TO MAKE IT Give physical resource tokens and transparent rules, constraints, and priorities.

DESIGN CHECK Is there a defensible solution and a reason to reject alternatives?

09 UV / INVISIBLE MESSAGE

LEARNING CONNECTION Evaluating overlooked details or receiving a discovery-based confirmation.

HOW IT WORKS After investigating visible evidence, a UV light reveals a clue or brief rationale.

HOW TO MAKE IT Use a UV-reactive pen and appropriate paper; supply a low-power light and print alternative.

DESIGN CHECK Test lighting and readability; do not aim lights at eyes; offer a print alternative.

10 PAPER-STRIP ASSEMBLY

LEARNING CONNECTION Prioritizing, sequencing, or ordering evidence according to rules.

HOW IT WORKS When players arrange strips correctly, letters along one edge reveal a word.

HOW TO MAKE IT Write short sortable records, place one letter on each, print, cut, and shuffle.

DESIGN CHECK Can players determine the sequence from the task, not from the letters?

11 TRANSPARENT OVERLAY

LEARNING CONNECTION Selecting relevant evidence and applying filtering or map criteria.

HOW IT WORKS Players align a clear sheet or window mask to expose selected information.

HOW TO MAKE IT Print a data page; make acetate marks or cut windows and add alignment cues.

DESIGN CHECK Are orientation and registration unambiguous? Offer a flat printed equivalent.

12 ACCORDION-FOLD REVEAL

LEARNING CONNECTION Sequencing, following directions, or interpreting a selected result.

HOW IT WORKS Folding along marked lines brings scattered shapes into a short message.

HOW TO MAKE IT Create a patterned sheet, define a short reveal, mark folds and a TOP arrow.

DESIGN CHECK Does the fold follow the intended reasoning? Test print direction and a flat alternative.

EARN THE REVEAL Make hidden messages the result of using relevant evidence, not a shortcut around it.

PHYSICAL PROPS + CODES

13-18 | Turn a valid decision into a tangible response or message.

13 ROTATING DECODER WHEEL

LEARNING CONNECTION Applying configuration rules or testing constrained settings.

HOW IT WORKS Turning aligned paper discs reveals a valid setting through a pointer or window.

HOW TO MAKE IT Print two circles on sturdy stock, attach with a brad, and add meaningful rules.

DESIGN CHECK Do learners use the criteria, or can they brute-force every position?

14 JIGSAW / PIECE ASSEMBLY

LEARNING CONNECTION Reconstructing a process, system, map, or evidence relationship.

HOW IT WORKS Players build a meaningful whole from fragments, tiles, or partial diagrams.

HOW TO MAKE IT Split a diagram into large pieces and include process or logic clues to guide placement.

DESIGN CHECK Would the shape alone solve it? Make reasoning essential and pieces manageable.

15 CARD-EDGE MESSAGE

LEARNING CONNECTION Sequencing or checking an evidence-based order.

HOW IT WORKS Correctly arranged cards produce a readable message along their edges.

HOW TO MAKE IT Mark aligned edges of cards in the intended order, then shuffle the deck.

DESIGN CHECK Provide sufficient ordering evidence and a consistent orientation marker.

16 WINDOW MASK / HIDDEN FLAP

LEARNING CONNECTION Selecting the right criteria and interpreting a resulting explanation.

HOW IT WORKS A flap or window reveals only information justified by an earlier decision.

HOW TO MAKE IT Cut safe windows or create a fold-over self-check on the back of the reference card.

DESIGN CHECK Make the reveal easy to find and readable; do not hide essential instructions.

17 SYMBOL / SUBSTITUTION CIPHER

LEARNING CONNECTION Interpreting a meaningful report once learners have earned the key.

HOW IT WORKS Players translate symbols or encoded letters to read an operational message.

HOW TO MAKE IT Supply a fair key or let learners earn it through a related learning decision.

DESIGN CHECK Is translation genuinely useful, or just an unrelated decoding gate?

18 ACROSTIC / INDEXED MESSAGE

LEARNING CONNECTION Identifying relevant lines and checking the selected evidence.

HOW IT WORKS Players select text by a rule, then read initials or indexed letters.

HOW TO MAKE IT Write short records and specify how to extract or order their letters.

DESIGN CHECK Give fair signposting: what should be selected, and in what order?

KEEP IT ACCESSIBLE Provide an equivalent route when a prop, color effect, or decoder cannot be used.

HUNTS + SHARED FINALES

19-24 | Use locations, team knowledge, and chapter results meaningfully.

19 RIDDLE / REBUS

LEARNING CONNECTION Interpreting concise clues, making a transition, or directing attention.

HOW IT WORKS Players solve wordplay or image-sound clues to identify an object or source.

HOW TO MAKE IT Use familiar vocabulary; test for multiple interpretations and cultural assumptions.

DESIGN CHECK Could an unfamiliar audience understand the clue without outside trivia?

20 RED-FILTER / VISUAL DECODE

LEARNING CONNECTION Revealing a checked result or directing attention to relevant data.

HOW IT WORKS A colored filter reduces visual noise and exposes a hidden pattern or label.

HOW TO MAKE IT Print layered colors, supply a simple filter, and create a black-and-white alternative.

DESIGN CHECK Does it work under room lighting, and can color-blind players access it?

21 MAP / LOCATION HUNT

LEARNING CONNECTION Wayfinding, finding resources, or navigating services and spaces.

HOW IT WORKS Players interpret a map, follow clues, and verify an intended location or service.

HOW TO MAKE IT Provide a clear map, accessible route choices, and a reliable check-in method.

DESIGN CHECK Are routes, mobility needs, and an indoor or seated equivalent considered?

22 SPLIT-INFORMATION ROLES

LEARNING CONNECTION Cross-team communication, handoffs, or integrating specialist knowledge.

HOW IT WORKS Each participant holds different evidence and must communicate to decide.

HOW TO MAKE IT Prepare role cards with complementary facts and a shared decision worksheet.

DESIGN CHECK Does everyone have a meaningful contribution, without bottlenecks?

23 QR / VIRTUAL LOCKBOX

LEARNING CONNECTION Self-checking, immediate feedback, or combining verified answers.

HOW IT WORKS Players submit a justified answer and receive a short confirmation or next clue.

HOW TO MAKE IT Use a small validated answer form and include a printed answer-check alternative.

DESIGN CHECK No phone should be mandatory; test links, errors, privacy, and connectivity.

24 META-PUZZLE FINALE

LEARNING CONNECTION Synthesizing findings from several related challenges.

HOW IT WORKS Each chapter provides an artifact that must be interpreted with others at the end.

HOW TO MAKE IT Design the final decision first, then make each chapter reward contribute meaningful data.

DESIGN CHECK Does the finale require synthesis, not merely collecting three tokens?

MEANINGFUL CONNECTIONS A satisfying finale makes teams combine insights, not just collect unrelated passwords.

PRINTED REVEALS THAT FEEL MAGICAL

Printable constructions with a clear learning reason.

Each recipe shows a buildable technique. Start with the learning decision and use the reveal to reinforce it.

01 STRIPS THAT REVEAL A WORD

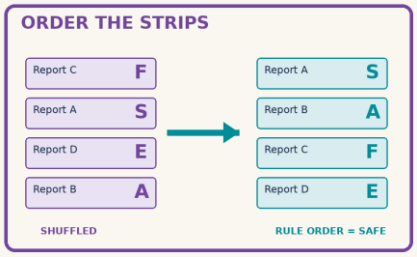
LEARNING CONNECTION Prioritize four incident reports using explicit escalation criteria.

HOW IT WORKS Players arrange paper strips in order. Letters in the edge column then spell SAFE.

HOW TO MAKE IT Write four reports and the ordering criteria; print one letter per strip in the intended order, cut and shuffle. Supply the decision guide and a reasoned answer flap.

DESIGN CHECK Can players establish the order from the criteria alone, without using SAFE to guess?

MATERIALS Printer, paper/cardstock, scissors, short guide



SCHEMATIC EXAMPLE · NOT TO SCALE

02 TRANSPARENT OVERLAY / WINDOW REVEAL

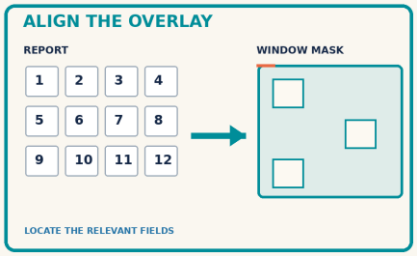
LEARNING CONNECTION Identify relevant details from a report or map using established rules.

HOW IT WORKS An aligned acetate layer or cutout sheet reveals the fields players should interpret.

HOW TO MAKE IT Design a report with scattered fields and a clear registration marker; print a transparent marked layer or window mask and label its orientation.

DESIGN CHECK Is there one obvious alignment? Can a player interpret the exposed data?

MATERIALS Printed data page, acetate or cutout paper, marker



SCHEMATIC EXAMPLE · NOT TO SCALE

MAKE IT EARNED Require the scenario decision before revealing the strip or overlay message. A visually neat result is not enough on its own.

ILLUMINATE, SPIN + DECIDE

Hands-on effects with accessible alternatives and verified outcomes.

03 UV INK / HIDDEN CONFIRMATION

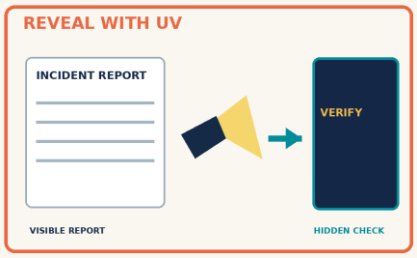
LEARNING CONNECTION Investigate evidence and check a safety or priority decision.

HOW IT WORKS After the decision, a flashlight reveals a brief confirmation or optional extra clue.

HOW TO MAKE IT Write with a UV-reactive pen on tested cardstock and provide an appropriate flashlight. Place the light at the right moment in the sequence.

DESIGN CHECK Test ink, lighting, batteries, and readability; include an equivalent printed reveal.

MATERIALS UV pen, low-power flashlight, paper and visible backup



SCHEMATIC EXAMPLE · NOT TO SCALE

04 ROTATING CALIBRATION WHEEL

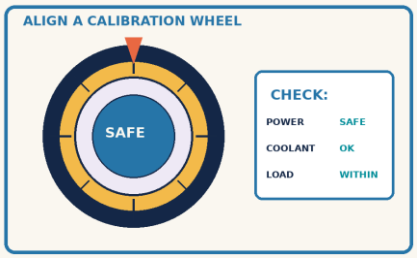
LEARNING CONNECTION Choose a safe configuration using technical constraints or allowable ranges.

HOW IT WORKS A spinning dial makes the selection physical; one marker shows the chosen alignment.

HOW TO MAKE IT Print two labeled concentric discs, cut and fasten with a brad. Supply equipment readings and rules that lead to one defensible setting.

DESIGN CHECK Can players brute-force the setting? Ask them to explain the rule before unlocking.

MATERIALS Cardstock, paper fastener/brad, printed constraints



SCHEMATIC EXAMPLE · NOT TO SCALE

TEST THE PHYSICAL EFFECT Check the UV light, batteries, and dial alignment in the actual room. Provide an equally informative printed alternative.

FOLD, COMBINE + SYNTHESIZE

Connected results that reward understanding, not guessing.

05 ACCORDION-FOLD MESSAGE

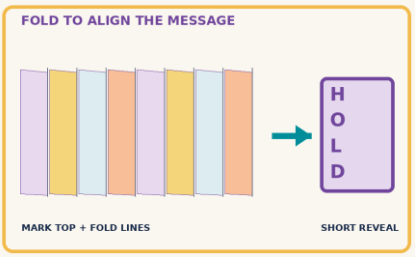
LEARNING CONNECTION Follow a meaningful sequence or reveal a result after analyzing information.

HOW IT WORKS A sheet of scattered shapes folds into a short word or symbol.

HOW TO MAKE IT Plan a brief reveal such as HOLD; arrange contrasting marks to align when folded. Print clear TOP and fold markers and test each fold.

DESIGN CHECK Do different printers change alignment? Offer a flat, readable alternative.

MATERIALS Paper, digital layout tool, fold guide



SCHEMATIC EXAMPLE · NOT TO SCALE

06 MEANINGFUL CONSTELLATION FINALE

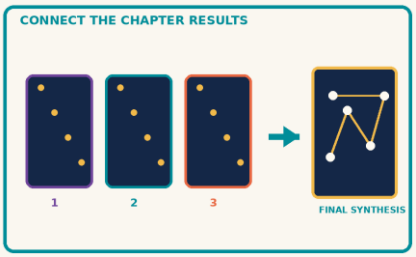
LEARNING CONNECTION Connect results from multiple challenges into a final shared task, then use the debrief to revisit the reasoning behind those results.

HOW IT WORKS Three recovered star-chart fragments combine into a larger map or conclusion.

HOW TO MAKE IT Plan the final reasoning step first. Give each chapter a chart tied to its result, and supply a marked assembly surface and short self-check.

DESIGN CHECK Could a team win by simply arranging symbols, without understanding earlier findings?

MATERIALS Three printed charts, assembly board, rationale check



SCHEMATIC EXAMPLE · NOT TO SCALE

CONNECT THE DISCOVERIES The final combination should use what earlier challenges taught players, not only ask them to collect disconnected symbols.

BEFORE YOU PRINT 50 COPIES

Test the learning decision, the game mechanics, and the learner experience.

A puzzle is ready when a fresh player can understand the goal, use the supplied evidence, reach a defensible outcome, and receive useful feedback. Apply all four checks below.

01 ALIGNMENT Does success require the target behavior? If players can bypass the reasoning, change the mechanic or checking method.

02 SOLVABILITY Is every required fact provided? Can a fresh tester reach the intended answer without the author explaining or prompting?

03 FACILITATION Are hints, rationale checks, time-up recovery, setup instructions, spare pieces, and reset steps ready?

04 ACCESSIBILITY Provide alternatives to tiny text, color-only patterns, audio/UV-only clues, obligatory phones, or hard-to-handle pieces.

A THREE-LEVEL HINT LADDER

1 NUDGE	2 FOCUS	3 RECOVER
Point to the goal, the relevant rule, or where to begin.	Name the specific evidence or constraint to inspect.	Demonstrate a first step; give a full reveal at time-up.

YOUR DESIGN MINI-BLUEPRINT

OBSERVABLE SUCCESS Given _____, learners will _____. Success means _____.

LEARNING MECHANIC To practice the decision, they will sort / compare / sequence / diagnose / _____.

REVEAL OR PROP The result will be confirmed through a strip / overlay / dial / self-check / _____.

FEEDBACK + TRANSFER The answer check explains _____. A new scenario will test _____.

NOVICE TEST Ask someone who did not create the puzzle to narrate their reasoning while playing. Record where the instructions, evidence, or feedback break down.

KEEP EXPLORING

Practical inspiration, attribution, and a quick method for adapting ideas.

These articles offer examples of puzzle construction and adventure design. Use them for inspiration; then create original scenarios, check accessibility, and verify that players practice your intended skill.

CONSTRUCTED ADVENTURES: HANDS-ON TECHNIQUES

↗ How to Build a Puzzle

Planning mechanics, parameters, clues, and playtesting.

↗ More Ways to Incorporate UV Blacklights

UV reveals, timing, audience expectations, and lighting pitfalls.

↗ Accordion Folding Paper Puzzle

A build process for folded, shape-based hidden messages.

↗ Crafting Puzzles Using a Deck of Cards

Card-order codes, marked cards, and edge reveals.

↗ Coffee-Shop Window Overlay

How a windowed sleeve can reveal information.

↗ Office Treasure-Hunt Puzzles

Ideas using everyday supplies and locations.

EDUCATIONAL ESCAPE-ROOM DESIGN GUIDANCE

↗ Room2Educ8: A Design-Thinking Framework for Educational Escape Rooms

An additional perspective on designing, prototyping, testing, and evaluating educational escape-room experiences.

ATTRIBUTION

Some graphic-novel character illustrations in this guide were generated using OpenAI image-generation tools and adapted for this resource. Drafting assistance was used for portions of the text and examples; all were edited and checked for instructional relevance.

Technique illustrations are original instructional schematics, not reproductions of third-party tutorials or commercial puzzle art.

MISSION ARCHITECT TOOLKIT / SESSION 606

AI-ASSISTED PUZZLE DESIGN

Ideas for using generative AI to assist with puzzle mechanics and creation.



Make sure to use approved tools and non-sensitive, fictional material and test everything yourself!

PROMPTS TO COPY AND ADAPT

01 IDEATE

Given [observable behavior], [audience], and [time/material constraints], propose three paper-based escape-room mechanics. For each, explain exactly what players decide, why that demonstrates the behavior, and what could go wrong.

02 DRAFT

Using ONLY [fictional scenario] and [supplied reference criteria], draft [number] clue or decision cards, one justified intended solution, two plausible distractors, and a simple self-check. Mark all assumptions and missing information.

03 CHALLENGE

Review these draft puzzle materials as a skeptical first-time player. Identify ambiguous clues, multiple valid solutions, rule violations, unintended shortcuts, and accessibility concerns. Flag what must be tested with real participants.

04 REFINE

Rewrite instructions in Goal → Do → Check format. Propose two increasingly specific hints, a time-up recovery path, and a debrief question that connects the decision to a new but comparable situation.

HUMAN VALIDATION: THE NON-NEGOTIABLE STEP

- ☐ Solve independently; verify every rule and the intended answer.
- ☐ Look for unsupported clues, contradictory constraints, and accidental shortcuts.
- ☐ Test the actual print pieces and hints with someone new.
- ☐ Check access, difficulty, and alignment before adding visual polish.

REAL DEVELOPMENT LESSON

During development of a workplace learning game, AI-generated logic-grid clues sometimes left multiple valid answers, while AI-drawn patrol routes violated movement rules. The designer revised the clues, drew correct routes, and independently verified the finished puzzles.

AI ATTRIBUTION Artwork generated with OpenAI tools; AI-assisted text reviewed and edited by the author.

PLAYTESTING & PUZZLE VALIDATION

Test the learning decision, puzzle logic, and player experience before launch.

Before you launch, test your activity with players to identify confusing instructions, ambiguous solutions, accessibility barriers, and unintended shortcuts. Use their feedback to revise and retest the experience.

BEFORE THE GAME

- ☐ Confirm the target learning behavior and what counts as success.
- ☐ Solve the puzzle yourself using only the information participants will receive.
- ☐ Check competing answers, ambiguous clues, dead ends, and shortcuts.
- ☐ Test fonts, contrast, reading level, manipulation, and non-tech options.
- ☐ Time reading, assembling, deciding, checking, and resetting puzzles separately.

DURING THE PLAYTEST

- ☐ Observe silently before intervening; record the exact point of confusion.
- ☐ Note whether everyone can meaningfully participate or one player dominates.
- ☐ Try self-checks and staged hints without relying on the facilitator.
- ☐ Give a three-minute and one-minute warning; test the time-up reveal.

AFTER THE PLAYTEST

- ☐ Ask: “Where were the instructions unclear? What caused you to get stuck? Did the feedback help you understand whether your solution was correct?”
- ☐ Separate challenge difficulty from confusing instructions or missing data.
- ☐ Revise, then test again with somebody new.

PLAYTEST OBSERVATION + REVISION LOG

WHAT HAPPENED?	Note the exact clue, behavior, or moment:
WHY?	Unclear instructions, missing data, unintended shortcut, access, or other:
WHAT WILL CHANGE?	Planned revision and owner:
RECHECK	Retest with a new solver after revisions; result:

MISSION ARCHITECT TOOLKIT / SESSION 606

LOGISTICS PLANNING

Practical planning for any room, team size, or facilitator setup.



____ attendees • ____ teams • ____ facilitator(s) • room setup: _____

CHOOSE A FORMAT THAT MATCHES THE REAL ROOM

Consider how many facilitators you have and what your bottlenecks might be. Rotating stations can make sense when movement supports the objective and you have space and staffing to manage it.

MATERIALS

Packets/kits, rulers, pencils, posters, etc. Include spares!

ROOM SETUP

Tables/chairs? Do you need rigid folders or lap boards?

ANSWER CHECK

How do they know they got the right answer? Hints?

PACING

How long should this take? Consider visible time cues.

EXPECT THE UNEXPECTED

What if the tech fails or they get stuck or don't finish?

INCLUSION

Make sure everyone can participate; roles, colors, layout, etc.

PROTECT COGNITIVE CAPACITY

Make the decision difficult, not the directions. State one clear goal, chunk materials by stage, use consistent visual cues, and watch whether novices struggle with the rules rather than the intended decision.

AI ATTRIBUTION Artwork generated with OpenAI tools; AI-assisted text reviewed and edited by the author.

MISSION ARCHITECT TOOLKIT / SESSION 606

DEBRIEF + LEARNING TRANSFER GUIDE

Turn game experiences into meaningful learning.

After learners play, help participants reflect on the decisions they made, explain their reasoning, and connect the experience to real-world skills. Consider a follow-up learning check to see if they can apply those skills.

A FIVE-MINUTE DEBRIEF

30 SEC RECALL / 2 MIN EXAMINE / 2 MIN APPLY / 30 SEC WRAP

Ask about the decision, not just the solution. Invite explanations, then connect the reasoning to another situation.

01 RECALL	**What happened?** "What did your team attempt? What new information changed your plan?"	**FACILITATOR MOVE** Invite a concise account; avoid a full clue-by-clue walkthrough.
02 EXAMINE	**Why did you decide that?** "Which evidence or constraint mattered most? What alternatives did you reject?"	**FACILITATOR MOVE** Probe the reasoning. Ask how they checked the result.
03 APPLY	**Where else would this work?** "Where might you make a similar decision? What would you need to know?"	**FACILITATOR MOVE** Connect to the real task and its limits; avoid a generic "teamwork" takeaway.

QUICK PROMPTS FROM THE BACKUP LEAGUE

MEMORY MELTDOWN

What made an incident genuinely consequential? Why did those three reports belong in that order?

GADGET GRAVEYARD

Which operating constraints ruled out a setting that initially looked workable?

ACADEMY CONFIDENTIAL

What evidence supported holding and verifying before deploying cadets?

IF A TEAM DID NOT FINISH

Reveal the story resolution without treating speed as ability. Ask what they attempted, which evidence they had, and what they would examine next. Unfinished play can still reveal valuable thinking.

IF YOU ONLY HAVE ONE MINUTE

Ask three fast questions: "What was the decision? Which evidence mattered? Where might that reasoning help again?" Capture one answer, then share the intended resolution.

CLOSE WITH THE SKILL "The rescue was the story. The question is how you made and checked decisions."

CHECK WHETHER LEARNING TRANSFERS

Plan evidence of learning without confusing team progress with individual mastery.

THREE LEVELS OF EVIDENCE

01 COMPLETION

What it shows: the team progressed through the game.

Check: final code, chapter result, or time to finish.

Not evidence of individual mastery.

02 REASONING

What it shows: someone can explain the choice using supplied criteria.

Check: ask each learner to justify the answer and reject an alternative.

Better evidence of understanding.

03 TRANSFER

What it shows: someone can apply the same decision process to a fresh case.

Check: assign a comparable scenario and score the response independently.

Stronger evidence of application.

PLAN AN (OPTIONAL) FOLLOW-UP LEARNING CHECK

After the debrief, consider giving each participant a short, unfamiliar scenario that requires the same reasoning to see if they can apply the new skill(s). Some questions to think through when planning the new interaction:

SKILL TO REUSE	What decision or action did the game help participants practice?
NEW SITUATION	What different scenario would require the same skill without allowing participants to repeat the memorized answer?
EVIDENCE OF LEARNING	What correct decision and explanation would demonstrate that someone understands the skill?
CHECK RESULTS	How and when will learners demonstrate the skill independently?

BACKUP LEAGUE EXAMPLE

Instead of HQ incident reports, give each learner six fictional maintenance requests with different safety consequences and timelines. Using a supplied, validated priority guide, ask them to select and rank the top three and explain the deciding evidence.

EVIDENCE OF LEARNING

☐ Decisions used the supplied criteria ☐ Learners explained their reasoning ☐ New-case performance checked

One thing to revise next time: _____

REMEMBER A correct final answer is satisfying feedback. It is not, by itself, a measure of individual performance.